Symmetry of time-bounded information from meta-complexity algorithms and average-case easiness

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Editorial note. Faithful editable TeX transcription of formal Theorem4.1 and author main Theorem1.2, complete Section4 proof pp26:15-26:16, exact necessary definitions and cited prior tools from Sections2.1/3, and complete published bibliography. Original author mathematical expressions and numbering retained. Standalone typography and long-equation line breaks adapted; no new proofs or mathematical corrections.

Allowed standard prerequisites (editorial).

- Approximation of time-bounded Kolmogorov complexity from errorless NP easiness. Exact author Lemma2.1 citing prior [31,33] retained.
- Deterministic direct-product reconstruction under an E circuit lower bound. Exact prior author Lemma2.2 citing [35], generator definition, and Fact3.4 retained.

3. Preliminaries: Notation. $[n]$ denotes $\{1,\dots,n\}$. For a function $p : \mathbb{N} \to \mathbb{N}$ and $i \in \mathbb{N}$, the function $p^{(i)} : \mathbb{N} \to \mathbb{N}$ is recursively defined as follows: $p^{(0)}(n) := n$ and $p^{(i+1)}(n) := p^{(i)}(p(n))$ for every $n \in \mathbb{N}$.

Kolmogorov Complexity. For a string $x \in \{0,1\}^*$ and an index $i \in \mathbb{N}$, let x_i denote the i-th bit of x if $i \leq |x|$ and $\perp$ if $i > |x|$. We fix an efficient universal Turing machine U. Time-bounded Kolmogorov complexity is formally defined as follows.

Definition 3.1 (Time-bounded Kolmogorov complexity). For strings $x, y \in \{0,1\}^*$, a time bound $t \in \mathbb{N} \cup \{\infty\}$, and an oracle A, the A-oracle t-time-bounded Kolmogorov complexity of x given y is defined as

$$K^{t,A}(x \mid y) := \min\{|d| : U^{A,y,d} \text{ outputs } x_i \text{ on input } i \text{ in time } t \text{ for every } i \in [1,\dots,|x|+1]\}.$$

Here, $U^{A,y,d}$ indicates that the universal Turing machine is given oracle access to A and each bit of y and $d \in \{0,1\}^*$. We omit the superscript A if $A = \varnothing$, the superscript t if $t = \infty$, and "$\mid y$" if y is the empty string. A simple counting argument implies the following basic fact of Kolmogorov-randomness.

Fact 3.2. For any integer $s \geq 1$ and any string $y \in \{0,1\}^*$, the number of strings $x \in \{0,1\}^*$ such that $K(x \mid y) < s$ is less than 2^s.

Proof. The number of programs of length less than s is at most $\sum_{i=0}^{s-1} 2^i < 2^s$. □

Definition 3.3 ([50,33]). For a polynomial $\tau : \mathbb{N} \to \mathbb{N}$, let

$$\Pi_{\text{Yes}} := \{(x, 1^t, 1^s) : K^t(x) \leq s\},$$

$$\Pi_{\text{No}} := \{(x, 1^t, 1^s) : K^{\tau(|x|,t)}(x) > s + \log \tau(|x|, t)\}.$$

We define $\mathsf{Gap}_\tau\mathsf{MINKT}$ to be the promise problem $(\Pi_{\text{Yes}}, \Pi_{\text{No}})$. We say that $\mathsf{GapMINKT} \in \mathsf{P}$ if there exists some polynomial τ such that $\mathsf{Gap}_\tau\mathsf{MINKT} \in \mathsf{P}$. The problem $\mathsf{GapMINKT}$ can be equivalently formulated as follows.

Fact 3.4 ([35]). The following are equivalent.

1. $\mathsf{GapMINKT} \in \mathsf{P}$.
2. There exist a polynomial-time algorithm $\widetilde{K}$ and a polynomial p such that

$$K^{p(t)}(x) - \log p(t) \leq \widetilde{K}(x, 1^t) \leq K^t(x)$$

for every string $x \in \{0,1\}^*$ and every integer $t \geq |x|$.

Definition 1.1. Symmetry of information for time-bounded Kolmogorov complexity (SoI) refers to the following hypothesis: There exists a polynomial p such that for any strings $x \in \{0,1\}^*$ and $y \in \{0,1\}^*$, for every $t \geq |x| + |y|$,

$$K^{p(t)}(x \mid y) + K^{p(t)}(y) - \log p(t) \leq K^t(x, y). \tag{SoI}$$

Throughout this paper, we refer to this hypothesis as SoI. The statement $\mathsf{DistNP} \subseteq \mathsf{AvgP}$ means that for every language L in NP and for every polynomial-time samplable distribution D, there exists an errorless heuristic scheme that solves L on inputs sampled from D; we refer the reader to the survey of Bogdanov and Trevisan [16] for background on average-case complexity theory.

Prior tools from Section 2.1. Our proof of Theorem 1.2 is based on the meta-complexity of time-bounded Kolmogorov complexity: We crucially use the fact that time-bounded Kolmogorov complexity can be approximated in Heuristica.

Lemma 2.1 ([31,33]). If $\mathsf{DistNP} \subseteq \mathsf{AvgP}$, then $\mathsf{GapMINKT} \in \mathsf{P}$. The second lemma is the reconstruction property of a k-wise direct product generator [34], which turned out to be a fundamental tool for analyzing Kolmogorov complexity [33,65,32,35]. A k-wise direct product generator $\mathsf{DP}_k : \{0,1\}^n \times \{0,1\}^{nk} \to \{0,1\}^{nk+k}$ is defined as follows:

$$\mathsf{DP}_k(x; z) := (z^1, \ldots, z^k, \langle z^1, x\rangle, \ldots, \langle z^k, x\rangle)$$

for every $x \in \{0,1\}^n$ and every $z = (z^1, \ldots, z^k) \in (\{0,1\}^n)^k$, where $\langle x, y\rangle$ denotes the inner product of x and $y \in \{0,1\}^n$ over GF(2), i.e., $\langle x, y\rangle := (\sum_{i=1}^n x_i y_i) \bmod 2$. The k-wise direct product generator $\mathsf{DP}_k(x; -)$ is a pseudo-random generator secure against an algorithm D if $K(x \mid D) > k + O(\log |x|)$. More formally, we have the following property:

Lemma 2.2 (Deterministic Reconstruction for DP_k; see [35]). Assume that $\mathsf{E} \not\subseteq \mathsf{i.o.SIZE}(2^{\epsilon n})$ for some constant $\epsilon > 0$. Then, there exists a polynomial p such that, for every $n \in \mathbb{N}$, $x \in \{0,1\}^n$, parameters $k, \epsilon^{-1}, s \in \mathbb{N}$, and for every randomized circuit D of size s such that

$$\left| \Pr_{z,r}[D(\mathsf{DP}_k(x; z); r) = 1] - \Pr_{w,r}[D(w; r) = 1] \right| \geq \epsilon,$$

where $z \sim \{0,1\}^{nk}$, $w \sim \{0,1\}^{nk+k}$, and $r \sim \{0,1\}^s$, it holds that

$$K^{p(ns/\epsilon)}(x \mid D) \leq k + \log p(ns/\epsilon).$$

The assumption of Lemma 2.2 is satisfied in Heuristica, as Buhrman et al. [18] showed that $\mathsf{E} \not\subseteq \mathsf{i.o.SIZE}(2^{\epsilon n})$ for some constant $\epsilon > 0$ if $\mathsf{DistNP} \subseteq \mathsf{AvgP}$. We mention in passing that the proof of Lemma 2.1 also uses Lemma 2.2.

Theorem 4.1. If $\mathsf{GapMINKT} \in \mathsf{P}$ and $\mathsf{E} \not\subseteq \mathsf{i.o.SIZE}(2^{\epsilon n})$ for some constant $\epsilon > 0$, then SoI holds.

Theorem 1.2. If $\mathsf{DistNP} \subseteq \mathsf{AvgP}$, then SoI holds.

4. Symmetry of Information from Average-Case Easiness of NP. In this section, we present a formal proof of Theorem 1.2. In fact, under a plausible assumption that E requires exponential-size circuits to compute, we prove that SoI follows from an approximation algorithm for time-bounded Kolmogorov complexity. Theorem 4.1 immediately implies Theorem 1.2.

Proof of Theorem 1.2. It is known that the two hypotheses of Theorem 4.1 are implied by the assumption that NP is easy on average: Buhrman et al. [18] showed that $\mathsf{E} \not\subseteq \bigcap_{\epsilon>0} \mathsf{i.o.SIZE}(2^{\epsilon n})$ if $\mathsf{DistNP} \subseteq \mathsf{AvgP}$; We also have $\mathsf{GapMINKT} \in \mathsf{P}$ if $\mathsf{DistNP} \subseteq \mathsf{AvgP}$ by Lemma 2.1. □

We now present the formal proof of Theorem 4.1.

Proof of Theorem 4.1. Let $\widetilde{K}$ be the polynomial-time algorithm of Fact 3.4 which satisfies the property that there exists a polynomial p such that for every $x \in \{0,1\}^*$ and every $t \geq |x|$,

$$K^{p(t)}(x) - \log p(t) \leq \widetilde{K}(x; 1^t) \leq K^t(x). \tag{9}$$

Fix strings $x \in \{0,1\}^n$ and $y \in \{0,1\}^m$ and an integer $t \geq n+m$. The proof of SoI is given by analyzing the following three values for $z \sim \{0,1\}^{nk}$, $w \sim \{0,1\}^{nk+k}$, $z' \sim \{0,1\}^{m\ell}$, and $w' \sim \{0,1\}^{m\ell+\ell}$:

$$\widetilde{K}(\mathsf{DP}_k(x;z), \mathsf{DP}_\ell(y;z'); 1^{t'}),$$
$$\widetilde{K}(w, \mathsf{DP}_\ell(y;z'); 1^{t'}),$$
$$\widetilde{K}(w, w'; 1^{t'}),$$

where $t' = t^{O(1)}$, $k \approx K^t(x,y) - \ell$, and $\ell \approx K^{\mathrm{poly}(t)}(y)$ are parameters chosen later. First, observe that Fact 3.2 implies that $K(w,w') \geq |w| + |w'| - 2$ with probability at least 3/4. Let $\theta := |w| + |w'| - 2 - \log p(t')$; using Equation (9), we obtain

$$\Pr_{w,w'}[\widetilde{K}(w,w';1^{t'}) \geq \theta] \geq \frac{3}{4}. \tag{10}$$

Next, we set the parameter ℓ to $K^{p'(t)}(y) - \log p'(t) - 1$, where p' is some large polynomial. Consider a randomized circuit D that takes w' as input as well as random bits w and outputs 1 if and only if $\widetilde{K}(w,w';1^{t'}) \geq \theta$. By the contrapositive of Lemma 2.2, $\mathsf{DP}_\ell(y;-)$ is a pseudorandom generator secure against D; i.e., D cannot distinguish $\mathsf{DP}_\ell(y;z')$ and w' in the sense that

$$\left| \Pr_{w,z'}[\widetilde{K}(w, \mathsf{DP}_\ell(y;z'); 1^{t'}) \geq \theta] - \Pr_{w,w'}[\widetilde{K}(w,w';1^{t'}) \geq \theta] \right| < \frac{1}{4},$$

which, together with Equation (10), implies that

$$\Pr_{w,z'}[\widetilde{K}(w, \mathsf{DP}_\ell(y;z'); 1^{t'}) \geq \theta] \geq \frac{1}{2}.$$

Finally, we compare $\widetilde{K}(w, \mathsf{DP}_\ell(y;z'); 1^{t'})$ with $\widetilde{K}(\mathsf{DP}_k(x;z), \mathsf{DP}_\ell(y;z'); 1^{t'})$. On one hand, since $|w| = |z| + k$ and $|w'| = |z'| + \ell$, we have

$$\Pr_{w,z'}[\widetilde{K}(w, \mathsf{DP}_\ell(y;z'); 1^{t'}) \geq |z| + |z'| + k + \ell - 2 - \log p(t')] \geq \frac{1}{2}. \tag{11}$$

On the other hand, observe that for some $t' := \mathrm{poly}(t)$,

$$\begin{aligned}\widetilde{K}(\mathsf{DP}_k(x;z), \mathsf{DP}_\ell(y;z'); 1^{t'}) &\leq K^{t'}(\mathsf{DP}_k(x;z), \mathsf{DP}_\ell(y;z')) \\ &\leq K^t(x,y) + |z| + |z'| + O(\log n)\end{aligned}$$

holds because the strings $\mathsf{DP}_k(x;z)$ and $\mathsf{DP}_\ell(y;z')$ can be computed from k, ℓ, z, z', and a program of size $K^t(x,y)$ that outputs (x,y) in time t. We now set $k := K^t(x,y) - \ell + O(\log t)$ so that

$$\Pr_{z,z'}[\widetilde{K}(\mathsf{DP}_k(x;z), \mathsf{DP}_\ell(y;z'); 1^{t'}) < |z| + |z'| + k + \ell - 2 - \log p(t')] = 1. \tag{12}$$

Let D_y be a randomized circuit that takes an input w and random bits z' and outputs 1 if and only if $\widetilde{K}(w, \mathsf{DP}_\ell(y;z'); 1^{t'}) < |z| + |z'| + k + \ell - 2 - \log p(t')$. It follows from Equations (11) and (12) that

$$\Pr_{z,z'}[D_y(\mathsf{DP}_k(x;z);z') = 1] - \Pr_{w,z'}[D_y(w;z') = 1] \geq 1 - \frac{1}{2} = \frac{1}{2}.$$

Using Lemma 2.2, we obtain

$$K^{\mathrm{poly}(t)}(x \mid D_y) \leq k + O(\log t) = K^t(x, y) - K^{p'(t)}(y) + O(\log t).$$

It follows that for some large polynomial q,

$$\begin{aligned} K^{q(t)}(x \mid y) &\leq K^t(x \mid D_y) + K^t(D_y \mid y) + O(1) \\ &\leq K^t(x, y) - K^{p'(t)}(y) + O(\log t) \\ &\leq K^t(x, y) - K^{q(t)}(y) + \log q(t) \end{aligned}$$

as desired. □

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